MS Algebra Recitation 1

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1 The Fundamental Group of a Topological Space

Let X be a topological space (if you don't know what that is, take X to be a metric space). A *path* in X is a continuous map $\gamma : [0,1] \to X$[1]. The endpoints of γ are $\gamma(0)$ and $\gamma(1)$. If $\gamma(0) = \gamma(1)$, we say that γ is a *loop* based at $\gamma(0)$.

Remark. Recall that the circle is given by $S^1 = [0,1]/\sim$, where $0 \sim 1$. We will often view loops as continuous maps $S^1 \to X$.

Remark. We will primarily be using paths to characterize spaces. As such, throughout these notes, all spaces are *path connected* (i.e. for any $x, y \in X$, there exists a path $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$).

We want to exaime how paths can be deformed into one another. This is the notion of a homotopy.

Definition. Let X be a topological space and $\gamma_0, \gamma_1 : [0,1] \to X$ be paths with $\gamma_0(0) = \gamma_1(0) = x$ and $\gamma_0(1) = \gamma_1(1) = y$. A (endpoint-preserving) *homotopy* from γ_0 to γ_1 is a (continuous) map $F : [0,1] \times [0,1] \to X$ such that:

- For all $s \in [0,1]$, $F(s,0) = \gamma_0(s)$ and $F(s,1) = \gamma_1(s)$
- For all $t \in [0,1]$, $F(0,t) = x$ and $F(1,t) = y$

We write $F_t(s) = F(s,t)$. If such an F exists, we say γ_0 and γ_1 are *homotopic*, and write $\gamma_0 \simeq \gamma_1$.

Intuitively, the variable t represents "time", and we are continuously deforming γ_0 into γ_1 without moving its endpoints.

Example. Let $X = \mathbb{R}^2$ and $\gamma_0(s) = 0$ be the constant loop. For any loop γ_1 based at 0, we have that $\gamma_0 \simeq \gamma_1$ via the homotopy
$$F_t(s) = t \cdot \gamma_1(s).$$

Now, for any paths $\gamma_0, \gamma_1 : [0,1] \to X$ with $\gamma_0(1) = \gamma_1(0)$, we define their *concatenation* as
$$\gamma_0 * \gamma_1(s) = \begin{cases} \gamma_0(2s) & 0 \le s \le 1/2 \\ \gamma_1(2s-1) & 1/2 \le s \le 1 \end{cases}$$

Proposition. *Homotopy forms an equivalence relation. Further, if $\gamma_0 \simeq \gamma_1$ and $\alpha_0 \simeq \alpha_1$ with $\gamma_i(1) = \alpha_i(0)$, then $\gamma_0 * \alpha_0 \simeq \gamma_1 * \alpha_1$.*

Proposition. *For paths $\gamma_0, \gamma_1, \gamma_2 : [0,1] \to X$ with $\gamma_i(1) = \gamma_{i+1}(0)$ ($i = 1,2$),*
$$(\gamma_0 * \gamma_1) * \gamma_2 \simeq \gamma_0 * (\gamma_1 * \gamma_2).$$

With this, we can define the fundamental group.

Definition. Let X be a topological space and $x_0 \in X$. The *fundamental group of X based at x_0*, denoted $\pi_1(X, x_0)$, is the set of equivalence classes of loops based at x_0 (under homotopy) with the binary operation $*$.

We have seen associativity. The identity is the trivial loop $s \mapsto x_0$, and the inverse of $\gamma(s)$ is $\overline{\gamma}(s) = \gamma(1-s)$.

Remark. The fundamental group doesn't depend on the base point x_0. For a path $\alpha : x_0 \to y_0$, we transform a loop γ based at y_0, we transform it into a loop based at x_0 given by $\alpha * \gamma * \overline{\alpha}$, and this gives an isomorphism $\pi_1(X, y_0) \to \pi_1(X, x_0)$.

Example. Since every loop in $\mathbb{R}^2$ is homotopic to the trivial loop, $\pi_1(\mathbb{R}^2)$ is trivial. More generally, $\pi_1(\mathbb{R}^d) = 0$.

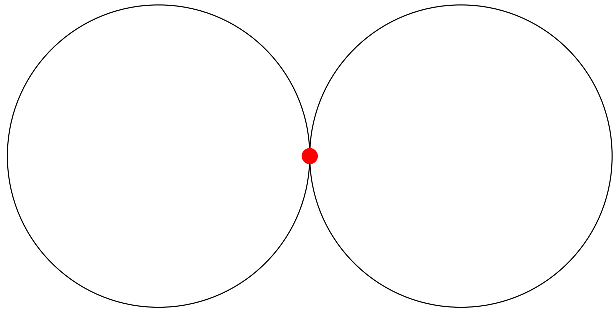
Example. $\pi_1(S^1) \cong \mathbb{Z}$, where a loop is mapped to the number of times it "winds" around the circle.

What we want to highlight is that the fundamental group is known as *functorial.* On top of giving us groups associated to topological spaces, it also gives us group homomorphisms associated to continuous maps. Suppose that $f : X \to Y$ is continuous, and fix points $x_0 \in X$ and $f(x_0) = y_0 \in Y$. We define $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ by $[\gamma] \mapsto [f \circ \gamma]$. It is not hard to check that f_* is a homomorphism and that $(g \circ f)_* = g_* \circ f_*$.

[1] Note this is strictly more information than the image of γ

2 Free Groups

What is the fundamental group of the following space?



We'll take x_0 to be the red point. There are two immediately obvious nontrivial loops: going around the left circle (call this α) and going around the right circle (call this β). This gives us loops $\overline{\alpha}$ and $\overline{\beta}$ for free, along with all possible concatinations of these four loops.

Are there any other loops? If we imagine the first time a loop returns to x_0, it must either come from the left circle or the right circle, and so it must start off going into that same circle. This means that the first part of the loop is homotopic to one of the trivial loop or $\alpha, \beta, \overline{\alpha}, \overline{\beta}$.

So every element in the fundamental group is some combination of $\alpha, \beta, \overline{\alpha}, \overline{\beta}$. Are there any nontrivial relations between them? For example, do we have $\alpha\beta = \beta\alpha$? Intuitively, we don't: the "first" direction we go in a loop determines whether its decomposition starts with $\alpha, \beta, \overline{\alpha}$, or $\overline{\beta}$ (and induction gives uniqueness). This is in fact the case. This group is what is known as a *free group.*

Generally, given a set A, we define A^{-1} to be a set $\{a^{-1} : a \in A\}$, where a^{-1} is a formal symbol such that $A \cap A^{-1} = \varnothing$. We want to consider words on $A \sqcup A^{-1}$: that is, sequences of symbols $a_1^{\pm} a_2^{\pm} \cdots a_n^{\pm}$ (we also allow for an empty word ϵ). we can juxtapose words

$$(a_1^{\pm} \cdots a_n^{\pm})(b_1^{\pm} \cdots b_m^{\pm}) = a_1^{\pm} \cdots a_n^{\pm} b_1^{\pm} \cdots b_m^{\pm}.$$

To make this into a group, we need inverses to work. So, we define an equivalence relation on words where we declare $AB \sim Aaa^{-1}B \sim Aa^{-1}aB$ (and take the equivalence relation generated by this). Just like with the fundamental group, juxtaposition plays nicely with this equivalence relation.

Definition. Let A be a set. The *free group on* A, denoted F_A, is the set of equivalence classes of words on $A \sqcup A^{-1}$ with the binary operation of juxtaposition.

So the fundamental group we encountered earlier was the free group on $\{\alpha, \beta\}$. Note that there is a natural inclusion $\iota : A \to F_A$ (where $a \in A$ maps to the one-letter word a).

Example. The free group on one generator is isomorphic to $\mathbb{Z}$.

Theorem 2.1 (Universal Property of Free Groups)**.** *Let G be a group and A a set, with some map $f : A \to G$. Then there is a unique map $F : F_A \to G$ with $F \circ \iota = f$. That is, the following diagram commutes:*

$$\begin{tikzcd} F_A \arrow[dr, dashed, "\exists! F"] & \\ A \arrow[u, "\iota"] \arrow[r, "f"'] & G \end{tikzcd}$$

This map is given by $a_1^{e_1} \cdots a_n^{e_n} \mapsto f(a_1)^{e_1} \cdots f(a_n)^{e_n}$.

Once again, we can lift maps of one form to maps of another. Given a set map $f : A \to B$, we get $F : F_A \to F_B$ with

$$\begin{tikzcd} F_A \arrow[r, dashed, "\exists! F"] & F_B \\ A \arrow[u, "\iota_A"] \arrow[r, "f"'] & B \arrow[u, "\iota_B"'] \end{tikzcd}$$

We get this by applying the universal property to $\iota_B \circ f : A \to F_B$.